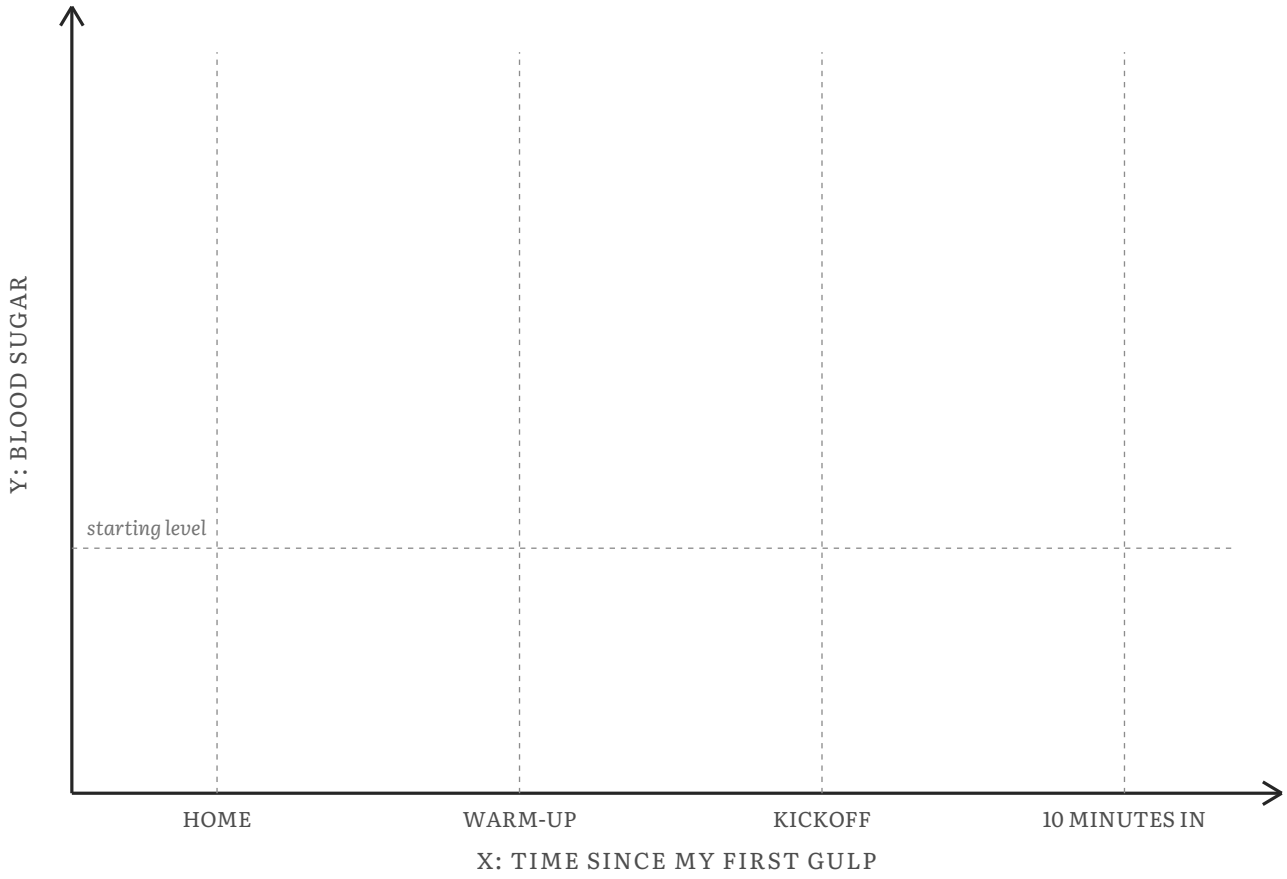
Draw the Roller Coaster

Read a line graph and connect it to the story.

Ages 7–9 · 15 min · Book link: Lab Notes, page 39

1. Look at Jiwoo's curve on page 39. Find HOME, WARM-UP, KICKOFF and 10 MINUTES IN.
2. On your sheet, draw Jiwoo's day. Write ELECTRIC where she felt amazing and WET SPAGHETTI where her legs gave up.
3. Now draw a second line in another color for a made-up day: same drink, but toast first. Label it MY GUESS.



☐ **Jiwoo's day** *(color 1)* ☐ **MY GUESS — toast first** *(color 2)*

Talk about it — why is your second line a guess and not a measurement?

A graph is a picture of an idea. Mine was a picture of one Saturday. — L.

The Insulin Crew Relay

Model how insulin and working muscles move sugar out of the blood — and why two grabbers at once empty it fast.

Ages 6–9 · 20–25 min · Book link: Chapter 6 and Lab Notes, pages 40–41

YOU NEED

20 paper "sugar cubes" (printable or crumpled paper), three boxes labeled BLOOD, CELLS and LEG MUSCLES, role cards, a timer. Signs, role cards and cubes are on the next pages.

HOW TO PLAY

1. Hand out roles: four Insulin Crew (IRIS, IGGY, INA, INKY), one Brain, two Leg Muscles, and everyone else as the Pancreas Alarm.
2. The rule for the Brain: there must be at least 2 cubes in BLOOD. If there are fewer, the Brain says "foggy…" and everyone wobbles like wet spaghetti for five seconds.
3. The Pancreas Alarm shouts ALARM! whenever BLOOD holds more than 5 cubes, and ALL CLEAR when it's back to 5 or fewer. The Crew may only carry cubes (one at a time, walking) from BLOOD to CELLS while the alarm is on.
4. Round 1 — toast and egg first. The grown-up drops 2 cubes into BLOOD every 10 seconds for one minute. Watch the alarm come and go.
5. Round 2 — three glasses, empty stomach. Dump 15 cubes in at once. Emergency mode: the Crew keeps carrying until the alarm team counts to 10 after ALL CLEAR. What happens to the Brain's cubes?
6. Round 3 — kickoff. Same dump, but now the Leg Muscles grab cubes from BLOOD too — they don't wait for the alarm. How fast does BLOOD empty now?
7. Talk: which round emptied BLOOD fastest? Who were the two grabbers? What could Jiwoo change first?

For the grown-up

It's a model, so it simplifies: insulin doesn't really carry sugar — it signals cells to take it in. (Lumi admits it on page 40: "The Crew is not really a crew.") If a player lives with diabetes, follow their care team's guidance and let them choose their role; never ask them to explain their condition.

Nobody in this game is the villain. Not the sugar, not the Crew, not your legs. — L.

Who's who in your body

IRIS

INSULIN CREW



Carry one cube at a time from BLOOD to CE only while the alarm is on. Walk, don't run

IGGY

INSULIN CREW



Carry one cube at a time from BLOOD to CE only while the alarm is on. Walk, don't run

INA

INSULIN CREW



Carry one cube at a time from BLOOD t only while the alarm is on. Walk, don't

INKY

INSULIN CREW



Carry one cube at a time from BLOOD to C only while the alarm is on. Walk, don't ru

BRAIN

ALWAYS ON DUTY

I need at least 2 cubes in BLOOD. Fewer? Say "foggy…" — and everybody wobbles like wet spaghetti for five seconds.

PANCREAS ALARM

EVERYONE ELSE

Shout ALARM! when BLOOD holds more than 5 cubes. Shout ALL CLEAR when it's 5 or fewer.

LEG MUSCLES

ROUND 3 ONLY

Grab cubes from BLOOD into LEG MUSCLES — you don't wait for the alarm.

LEG MUSCLES

ROUND 3 ONLY

Grab cubes from BLOOD into LEG MUSCLES — you don't wait for the alarm.

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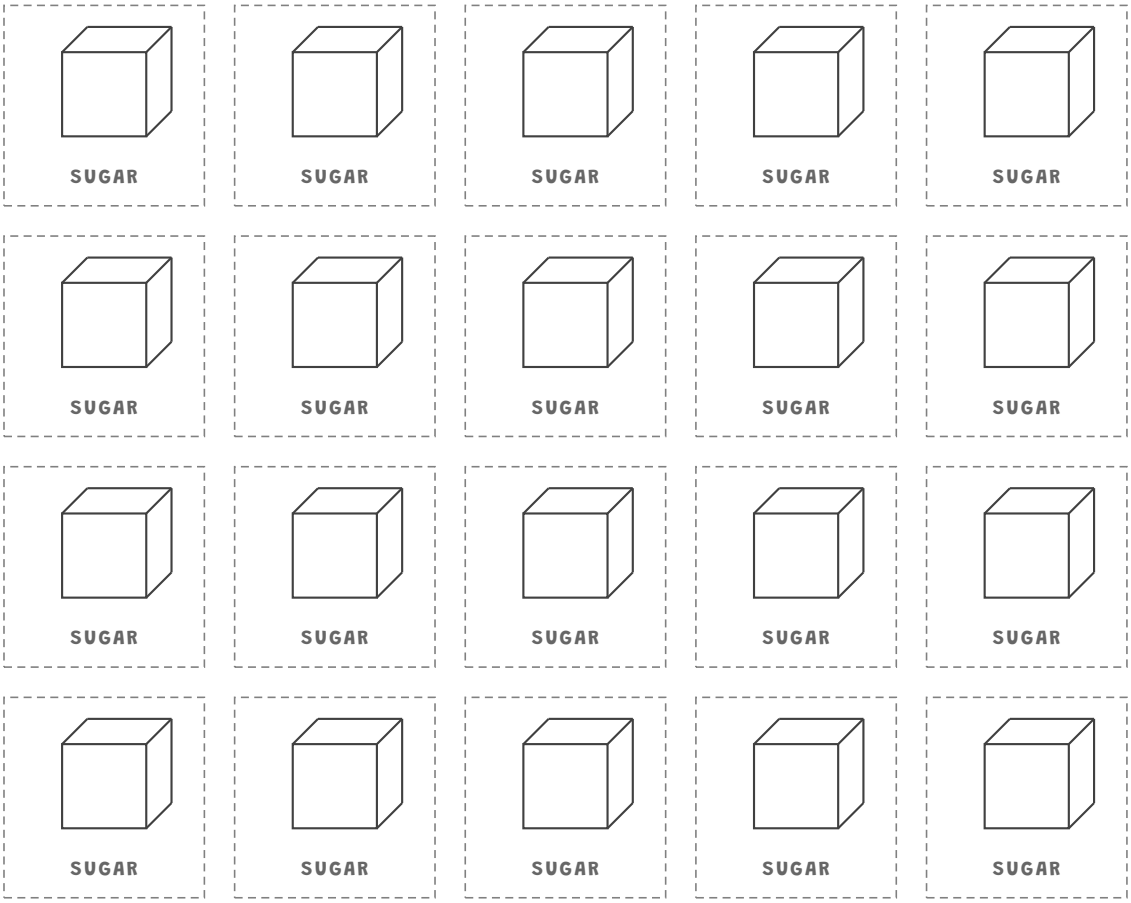
Signs for the boxes

BLOOD

CELLS

LEG MUSCLES

20 SUGAR CUBES



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Four Things Were True at Once

Tell causes from coincidences, then plan a fair test that changes one thing.

Ages 7–9 · 15–20 min · Book link: Lab Notes, pages 42–43

1. Cut out the twelve cards. Sort them onto the mat: Part of the crash or Just true that day.
2. Check your sort against Lumi's list on page 42. Did you find all four?

On an empty stomach	**The drink was neon blue**	**A lot of sugar**
A cartoon hero was on the label	**Maya was stretching by the goal**	**It was Saturday**
All at once	**Right before sprinting**	**Theo brought orange slices**
It cost three weeks of trash money	**Jiwoo got there one minute late**	**The bottle had a trophy-cup lid**

Cut on the dashed lines.

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Part of the crash — or just true that day?

PART OF THE CRASH	JUST TRUE THAT DAY

Look at the hero card again. The cartoon on the label didn't change Jiwoo's blood sugar — but did it change what she did?

MY FAIR TEST

I will change: _____

I will keep the same: _____

I will notice: _____

Change all four and you learn nothing. Change one and you learn one thing. That's a lot. — L.

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Label Detective: FOR CHAMPIONS ONLY

Notice what a label or ad promises — and how a scientist would test the promise.

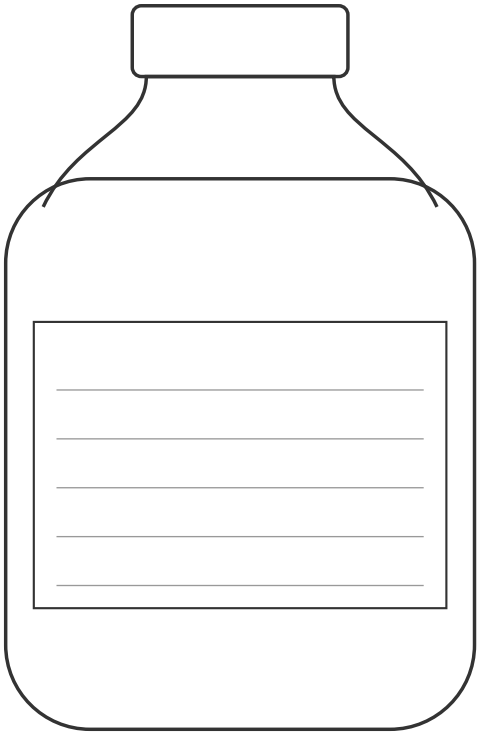
Ages 7–9 · 20 min · Book link: Chapter 1 and Think Like a Scientist, page 47 (question 1)

Lightning Kick's label



Sample size: one kid, one day, one drink.

Your honest label



1. What does Lightning Kick's label promise?

2. Who is on the label — and why do you think they put him there?

3. What does the label NOT tell you?

4. How could you test the promise fairly?

Grown-ups: This activity is about promises, not numbers. Please don't turn it into counting grams — the book leaves numbers out on purpose.

Lightning Kick is not the same thing as Lightning Kick's drink. That one took me a long time to learn. — L.

Lumi Shows Up

Write in Lumi's lab-report voice and retell an ordinary moment like a scientist.

Ages 6–9 · 15–20 min · Book link: Chapters 2–4

1 · Lumi pops out of…

2 · I was about to…

3 · Lumi writes on her clipboard:

"Subject reports: _____

4 · Later, I noticed…

Lumi's style: short, calm, a little dry. "Subject reports: electric." · "Subject ignored advice. Recording." · "On an empty stomach. Before a match. Bold."

I never shout. Well — once. Page 8. It was an emergency. — L.

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Word Lab Workout

Practice the words a scientist needs, from Lumi's Word Lab.

Ages 7–9 · 15 min · Book link: Lumi's Word Lab, pages 48–57

1 · FIND ALL 17 WORDS (across, down or diagonal)

B	S	D	A	R	H	E	E	O	P	M	B	K	B
H	C	U	D	B	G	M	N	B	V	U	G	C	L
S	A	C	G	M	S	P	E	S	F	P	S	A	O
U	R	I	V	A	I	T	R	E	Y	A	U	S	O
B	B	N	P	D	R	Y	G	R	F	W	G	E	D
J	O	S	A	G	V	S	Y	V	U	V	A	S	S
E	H	U	N	L	M	T	P	E	E	G	R	T	U
C	Y	L	C	U	B	O	M	I	L	L	C	U	G
T	D	I	R	C	E	M	B	G	K	I	R	D	A
B	R	N	E	O	D	A	T	A	V	E	A	Y	R
W	A	G	A	S	A	C	C	E	L	L	S	L	O
Y	T	M	S	E	F	H	V	K	C	G	H	B	R
T	E	R	C	H	Y	P	O	T	H	E	S	I	S
O	E	X	P	E	R	I	M	E	N	T	D	N	Y

- ☐ CASE STUDY
- ☐ SUBJECT
- ☐ DATA
- ☐ OBSERVE
- ☐ HYPOTHESIS
- ☐ EXPERIMENT
- ☐ ENERGY
- ☐ FUEL
- ☐ CARBOHYDRATE
- ☐ GLUCOSE
- ☐ BLOOD SUGAR
- ☐ EMPTY STOMACH
- ☐ SUGAR SPIKE
- ☐ PANCREAS
- ☐ INSULIN
- ☐ CELL
- ☐ SUGAR CRASH

2 · MATCH THE WORD TO LUMI'S DEFINITION

DATA	☐	A A quiet organ behind your stomach. Its job is to keep your blood sugar just right.
OBSERVE	☐	B A test you design to find out if something is true. You change one thing at a time.
HYPOTHESIS	☐	C To watch something carefully and notice what actually happens, without trying to change it.
EXPERIMENT	☐	D Information you collect about something. Pictures, measurements, notes, feelings — it's all data.
EMPTY STOMACH	☐	E A hormone your body makes to move glucose from your blood into your cells.
SUGAR SPIKE	☐	F A guess you make before you test something — based on what you know, but not proven yet.
PANCREAS	☐	G When a pile of sugar enters your blood all at once and your blood sugar shoots up like a rocket.
INSULIN	☐	H A stomach with no food in it.

3 · SAY IT LIKE A DOCTOR: *hypoglycemia — hy-po-gly-SEE-mee-uh*

Learn these once and you'll never need me to explain them again. I'd rather be collecting data anyway. — L.

Word Lab Workout — answers

For grown-ups and for checking your own work afterwards.

B	S	D	A	R	H	E	E	O	P	M	B	K	B
H	C	U	D	B	G	M	N	B	V	U	G	C	L
S	A	C	G	M	S	P	E	S	F	P	S	A	O
U	R	I	V	A	I	T	R	E	Y	A	U	S	O
B	B	N	P	D	R	Y	G	R	F	W	G	E	D
J	O	S	A	G	V	S	Y	V	U	V	A	S	S
E	H	U	N	L	M	T	P	E	E	G	R	T	U
C	Y	L	C	U	B	O	M	I	L	L	C	U	G
T	D	I	R	C	E	M	B	G	K	I	R	D	A
B	R	N	E	O	D	A	T	A	V	E	A	Y	R
W	A	G	A	S	A	C	C	E	L	L	S	L	O
Y	T	M	S	E	F	H	V	K	C	G	H	B	R
T	E	R	C	H	Y	P	O	T	H	E	S	I	S
O	E	X	P	E	R	I	M	E	N	T	D	N	Y

MATCHING

DATA — D

OBSERVE — C

HYPOTHESIS — F

EXPERIMENT — B

EMPTY STOMACH — H

SUGAR SPIKE — G

PANCREAS — A

INSULIN — E

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Junior Scientist Badge

Celebrate thinking like a scientist — whatever the result.

Ages 6–9 · 10 min · Book link: Think Like a Scientist, page 46



Scientist: _____

Lab partner: _____

Date: _____

This badge is for writing down what really happened. Even when nothing did. Especially then. — L.